

ADVANCED TECHNOLOGY & SUSTAINABLE ENERGY GENERATION: ROBOTICS TO POWER SOLAR OPERATIONS

In an effort to mitigate the effects of climate change, countries across the globe are adopting and investing in sustainable practices. Most nations have set ambitious, timebound targets for renewable energy capacity and are working towards achieving them. For India, which has set a goal of 500 GW by 2030, solar energy has emerged as a dominant force in the renewables sector. However, the process of generating energy through solar panels poses a challenge to sustainable practices. This

is because solar panels require regular cleaning and maintenance to maximise efficiency and increase power output but cleaning them is a water intensive process. Each solar panel needs 0.8-1.2 litres of water in most areas and 1-1.5 litres in dry regions per cycle. This could mean that a 1 MW solar farm will use up about 5,000 liters of water per cycle. With an aim to lower the water footprint of its solar portfolio, Adani Green Energy Limited (AGEL), which is one of the largest renewable companies in India, is

deploying a water-free robotic cleaning system. AGEL plans to implement this advanced technology across its operational solar and hybrid sites, and also its upcoming projects – especially in dry and arid regions of the country like Rajasthan and Gujarat, where renewable water sources are scarce. This will bring down water usage for solar module maintenance to zero.

AGEL is on its way to cover 4,830 MW of its solar portfolio with robotic cleaning. The move will save about 595 million

